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1.1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:
$$p = m \times v$$

where:
 m is the mass of the object (in kilograms).
 v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1 m=float(input(""))
2 v=float(input(""))
3 p=m*v
4 print(f"{p:.2f}kgm/s")
```

Average time0.015 s15.50 msMaximum time0.031 s31.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 143 msDebug

Expected output
5.0
18.0
50.00kgm/s

Actual output
5.0
18.0
50.00kgm/s

Test case 219 ms

TerminatTest Cases

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Breaking news
Vijay gets gover...

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9:46 PM
5/9/2026

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2.2.2. Captain of the Team

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You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

Input Format:
The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

Output Format:
The output should be the height (in centimeters) of the tallest player.

Sample Test Cases

captainof...

1heights=list(map(int,input()).split())
2tallest_player=max(heights)
3print(tallest_player)

Average time0.006 s5.47 ms
Maximum time0.008 s8.03 ms

1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1
Expected output: 171 180 185 190 174 195 188 198 187 177 198
171
Actual output: 171 180 185 190 174 195 188 198 187 177 198
171

TerminorTest cases

Show hidden icons

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1.2.4. Pattern - 2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 n=int(input())
2 for i in range(1,n+1):
3     for j in range(1,i+1):
4         print(j,end=" ")
5     print()
6
```

Average time0.010 s10.25 ms

Maximum time0.015 s15.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

1-
1 2-
1 2 3-
1 2 3 4-
1 2 3 4 5-

Actual output

1-
1 2-
1 2 3-
1 2 3 4-
1 2 3 4 5-

Terminal Test Cases

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3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations
 - Compute the element-wise sum, difference, and product of the two arrays.
2. Statistical Operations
 - Calculate the mean, median, and standard deviation of array A.
3. Bitwise Operations
 - Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A, OR B).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

different...

```

1 import numpy as np
2
3 def array_operations(A, B):
4
5     # Convert A and B to Numpy arrays
6     A = np.array(A, dtype=int)
7     B = np.array(B, dtype=int)
8
9     # Arithmetic Operations
10    sum_result = A + B
11    diff_result = A - B
12    prod_result = A * B
13
14    # Statistical Operations
15    mean_A = np.mean(A)
16    median_A = np.median(A)
17    std_dev_A = np.std(A)
18
19    # Bitwise Operations
20    and_result = A & B
21    or_result = A | B
22    xor_result = A ^ B
23
24    # Output results with one space between each element
25    print("Element-wise Sum:", " ".join(map(str, sum_result)))
26    print("Element-wise Differences:", " ".join(map(str, diff_result)))
27    print("Element-wise Products:", " ".join(map(str, prod_result)))
28
29    print(f"Mean of A: {mean_A}")
30    print(f"Median of A: {median_A}")
31    print(f"Standard Deviation of A: {std_dev_A}")
32
33    print("Bitwise AND:", " ".join(map(str, and_result)))
34    print("Bitwise OR:", " ".join(map(str, or_result)))
35    print("Bitwise XOR:", " ".join(map(str, xor_result)))

```

Average time

0.013 s

11.06 ms

Maximum time

0.020 s

20.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

4 8 16 12

-4 -4 -4 -4

8 12 12 12

Mean of A: 2.5

Actual output

4 8 16 12

-4 -4 -4 -4

8 12 12 12

Mean of A: 2.5

Test cases

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5.1.1. Stacked Plot

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

```

1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 # Data for Months and Temperature for three cities
5 data = {
6     'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August', 'September', 'October',
7              'November', 'December'],
8     'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 4],
9     'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
10    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
11 }
12
13 # Write your code...
14 df = pd.DataFrame(data)
15
16 plt.stackplot(df['Month'], df['City_A_Temperature'], df['City_B_Temperature'], df['City_C_Temperature'])
17
18 plt.xlabel('Month')
19 plt.ylabel('Temperature')
20 plt.title('Temperature Variation')
21 plt.legend(loc='upper left')
22 plt.show()

```

Average time
0.379 s
379.68 ms

Maximum time
0.379 s
379.68 ms

1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

Test cases

1 Pass

Result

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4.2.2. Best Selling Product

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	6	20	Los Angeles
2025-01-04	Product C	8	30	New York
2025-01-04	Product B	9	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 # Import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9
10 # Display the result
11 print(f"Best selling product: {best_product}")
12 print(f"Total quantity sold: {highest_quantity}")
13
14 # Prompt the user for the file name
15 file_name = input()
16
17 # Load the data
18 df = pd.read_csv(file_name)
19
20
21 # Find the product with the highest total quantity sold
22 best_product = df.groupby('Product')['Quantity'].sum().idxmax()
23 highest_quantity = df.groupby('Product')['Quantity'].sum().max()
24
25 # Display the result
26 print(f"Best selling product: {best_product}")
27 print(f"Total quantity sold: {highest_quantity}")
28
```

Average time: 0.022 s
Shortest time: 0.043 s

1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 43 ms Delay

Expected output:

```
SALES_DATA.csv
Best selling product: Product A
Total quantity sold: 25
```

Actual output:

```
SALES_DATA.csv
Best selling product: Product A
Total quantity sold: 25
```

Terminal Test cases

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1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 def fibonacci(n):
2     if n <= 1:
3         return n
4     elif n==2:
5         return 1
6     else:
7         return fibonacci(n - 1) + fibonacci(n - 2)
8
9
10 n = int(input())
11 for i in range(1, n + 1):
12     print(fibonacci(i), end=" ")
13
```

Average time
0.022 s
22.50 ms

Maximum time
0.067 s
67.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1

Expected output
0 1 1 2 3

Actual output
0 1 1 2 3

Test case 2

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY-MM-DD`.
- The second line contains the second date in the format `YYYY-MM-DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date

Sample Test Cases

noOfDays...

```
1 from datetime import date
2 from datetime import datetime
3 d1 = input().strip()
4 d2 = input().strip()
5
6 date1 = datetime.strptime(d1, "%Y-%m-%d")
7 date2 = datetime.strptime(d2, "%Y-%m-%d")
8
9 diff = date2 - date1
10 print(diff.days)
```

Average time0.014 s14.20 msMaximum time0.038 s38.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1

Expected output

Actual output

2014-07-042014-07-04

2014-11-022014-11-02

123123

Test case 2

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4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex)

2. Get the number of non-survivors by gender (Sex)

3. Get the number of survivors by embarkation location (Embarked_S)

4. Get the number of non survivors by embarkation location (Embarked_S)

5. Calculate the percentage of children (Age < 18) who survived

6. Calculate the percentage of adults (Age >= 18) who survived

7. Get the median age of survivors

8. Get the median age of non-survivors

9. Get the median fare of survivors

10. Get the median fare of non-survivors

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	3	S
2	1	1	Mr. Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17596	53.1	1	C85
3	1	3	Mr. Allou, Mr. William Henry	male	29	0	0	STON/O2	52.00	3	S
4	1	1	Mr. Fretwell, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	11380	51.1	1	C123
5	0	3	Mr. Niles, Mr. William Henry	male	29	0	0	STON/O2	52.00	3	S
6	0	3	Mr. Niles, Mr. James	male	34	0	0	STON/O2	52.00	3	S
7	0	3	Mr. McCarthy, Mr. Timothy J	male	34	0	0	STON/O2	52.00	3	S
8	0	3	Mr. Niles, Mr. James	male	34	0	0	STON/O2	52.00	3	S
9	1	1	Mr. Fretwell, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	11380	51.1	1	C123
10	1	1	Mr. Fretwell, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	11380	51.1	1	C123

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,Mr. Braund, Mr. Owen Harris,male,22,1,0,A/5 21171,7.25,S

2,1,1,Mr. Cumings, Mrs. John Bradley (Florence Briggs Thayer),female,38,1,0,PC 17596,53.1,C85,C

3,1,3,Mr. Allou, Mr. William Henry,male,29,0,0,STON/O2,52.00,S

4,1,1,Mr. Fretwell, Mrs. Jacques Heath (Lily May Peel),female,35,1,0,11380,51.1,C123,S

5,0,3,Mr. Niles, Mr. William Henry,male,29,0,0,STON/O2,52.00,S

6,0,3,Mr. Niles, Mr. James,male,34,0,0,STON/O2,52.00,S

7,0,3,Mr. McCarthy, Mr. Timothy J,male,34,0,0,STON/O2,52.00,S

8,0,3,Mr. Niles, Mr. James,male,34,0,0,STON/O2,52.00,S

9,1,1,Mr. Fretwell, Mrs. Jacques Heath (Lily May Peel),female,35,1,0,11380,51.1,C123,S

10,1,1,Mr. Fretwell, Mrs. Jacques Heath (Lily May Peel),female,35,1,0,11380,51.1,C123,S

Note: Refer to the visible test case for better reference.

Sample Test Cases

Static Data

```
1 # Import pandas as pd
2 # Import numpy as np
3
4 # Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
7 import pandas as pd
8 import numpy as np
9
10 # Load the Titanic dataset
11 data = pd.read_csv('Titanic-Dataset.csv')
12 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
13 # Load the Titanic dataset
14 data = pd.read_csv('Titanic-Dataset.csv')
15 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
16 # 1. Get the number of survivors by gender
17 survivors_by_gender=data[data['Survived']==1]['Sex'].value_counts()
18 print(survivors_by_gender)
19 # 2. Get the number of non-survivors by gender
20 non_survivors_by_gender=data[data['Survived']==0]['Sex'].value_counts()
21 print(non_survivors_by_gender)
22 # 3. Get the number of survivors by embarked location
23 survivors_by_embarked=data[data['Survived']==1]['Embarked_S'].value_counts()
24 print(survivors_by_embarked)
25 # 4. Get the number of non-survivors by embarked location
26 non_survivors_by_embarked=data[data['Survived']==0]['Embarked_S'].value_counts()
27 print(non_survivors_by_embarked)
28 # 5. Calculate the percentage of children (Age < 18) who survived
29 children_survival=data[data['Age']<18]['Survived'].mean()
30 print(children_survival)
31 # 6. Calculate the percentage of adults (Age >= 18) who survived
32 adults_survival=data[data['Age']>=18]['Survived'].mean()
33 print(adults_survival)
34 # 7. Get the median age of survivors
```

Average time: 0.109 s

Maximum time: 0.109 s

1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

Female --- 233

Female --- 233

male --- 180

male --- 180

Name: Sex, dtype: object

Name: Sex, dtype: object

Female --- 488

Female --- 488

Name: Sex, dtype: object

Name: Sex, dtype: object

Test cases

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6.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using `pd.get_dummies()`), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "Survival by Embarked" to the chart.
4. Label the x-axis as 'Embarked' and the y-axis as 'Count'.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as 'Survived' and 'Not Survived').

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,350877,8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,349909,21.079,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S
10,1,2,"Bassett, Mrs. Nicholas (Adeline Acham)",female,14,1,0,237736,20.0708,C

```

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf...

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Embarked
17 survival_counts = data.groupby('Embarked_Q')[['Survived']].value_counts().unstack().fillna(0)
18
19 survival_counts.plot(kind='bar', stacked=True)
20 plt.title('Survival by Embarked')
21 plt.xlabel('Embarked')
22 plt.ylabel('Count')
23 plt.legend(['Not Survived', 'Survived'])
24 plt.show()

```

Average time

0.414 s

Maximum time

0.414 s

1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

Survival by Embarked

Survival by Embarked

Terminat

Test cases

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4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Calculate the survival rate by class.
- Calculate the survival rate by embarked location (Embarked_S).
- Calculate the survival rate by family size (FamilySize).
- Calculate the survival rate by being alone (IsAlone).
- Get the average fare by passenger class (Pclass).
- Get the average age by passenger class (Pclass).
- Get the average age by survival status (Survived).
- Get the average fare by survival status (Survived).
- Get the number of survivors by class (Pclass).
- Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Brandy, Mr. Owen Harris", male, 26, 0, 0, 0, 517, 5.42, , S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, 0, 7534, 53.1, C85, C
3,1,3,"Heikkinen, Miss. Laina", female, 26, 0, 0, 0, 5101, 21.015, , S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 1, 11380, 53.1, C123, S
5,0,3,"Allen, Mr. William Henry", male, 35, 0, 0, 0, 37366, 8.66, , Q
6,0,3,"Moran, Mr. James", male, 0, 0, 0, 0, 5167, 5.42, , Q
7,0,1,"McCarthy, Mr. Timothy J", male, 54, 0, 0, 0, 51743, 51.8625, E46, S
8,0,2,"Palsson, Master. Gosta Leonard", male, 2, 1, 1, 0, 16000, 26.45, , S
9,1,1,"Townsley, Mrs. Clara (Elizabeth Vilhelmina Berg)", female, 27, 0, 2, 0, 14742, 51.8625, , S
10,1,2,"Nesher, Mrs. Nicolas (Adele Achen)", female, 14, 1, 0, 0, 12773, 20.6708, , C

```

Note: Refer to the visible test case for better reference.

Sample Test Cases

TitanicDat...

```

1 # Import pandas as pd
2 # Import numpy as np
3
4 # Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6 # data['FamilySize'] = data['SibSp'] + data['Parch']
7 # data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
8 # data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10 import pandas as pd
11 import numpy as np
12
13 # Load the Titanic dataset
14 data = pd.read_csv('Titanic-Dataset.csv')
15 data['FamilySize'] = data['SibSp'] + data['Parch']
16 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
17 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
18
19 # Load the Titanic dataset
20 data = pd.read_csv('Titanic-Dataset.csv')
21 data['FamilySize'] = data['SibSp'] + data['Parch']
22 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
23 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
24
25 # 1. Calculate the survival rate by class
26 survival_by_class = data.groupby('Pclass')['Survived'].mean()
27
28 # 2. Calculate the survival rate by embarked location
29 survival_by_embarked = data.groupby('Embarked_S')['Survived'].mean()
30
31 # 3. Calculate the survival rate by family size
32 survival_by_family = data.groupby('FamilySize')['Survived'].mean().sort_index()
33
34 # 4. Calculate the survival rate by being alone
35 survival_by_alone = data.groupby('IsAlone')['Survived'].mean()
36
37 # 5. Get the average fare by class
38 fare_by_class = data.groupby('Pclass')['Fare'].mean()

```

Average time: 0.111 s

Maximum time: 0.111 s

1 out of 1 shown test case(s) passed

Test Case 1

Expected output

Actual output

Pclass

1 → 0.629639

2 → 0.472629

3 → 0.242181

Pclass

1 → 0.629639

2 → 0.472629

3 → 0.242181

Name: Survived, dtype: float64

Embarked_S

Name: Survived, dtype: float64

Embarked_S

Test cases

Test cases

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4.2.3 City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	1	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	6	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	1	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthfor...

```
1 # Import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9
10 # Display the result
11 print(f"City sold the most products: {best_city}")
12 import pandas as pd
13
14 # Prompt the user for the file name
15 file_name = input()
16
17 # Load the data
18 df = pd.read_csv(file_name)
19
20 # write the code..
21 city_sales = df.groupby('City')['Quantity'].sum().reset_index()
22 best_city_row = city_sales.loc[city_sales['Quantity'].idxmax()]
23 best_city = best_city_row['City']
24 # Display the result
25 print(f"City sold the most products: {best_city}")
26
```

Average time: 0.019 s

Maximum time: 0.038 s

19.33 ms

38.88 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

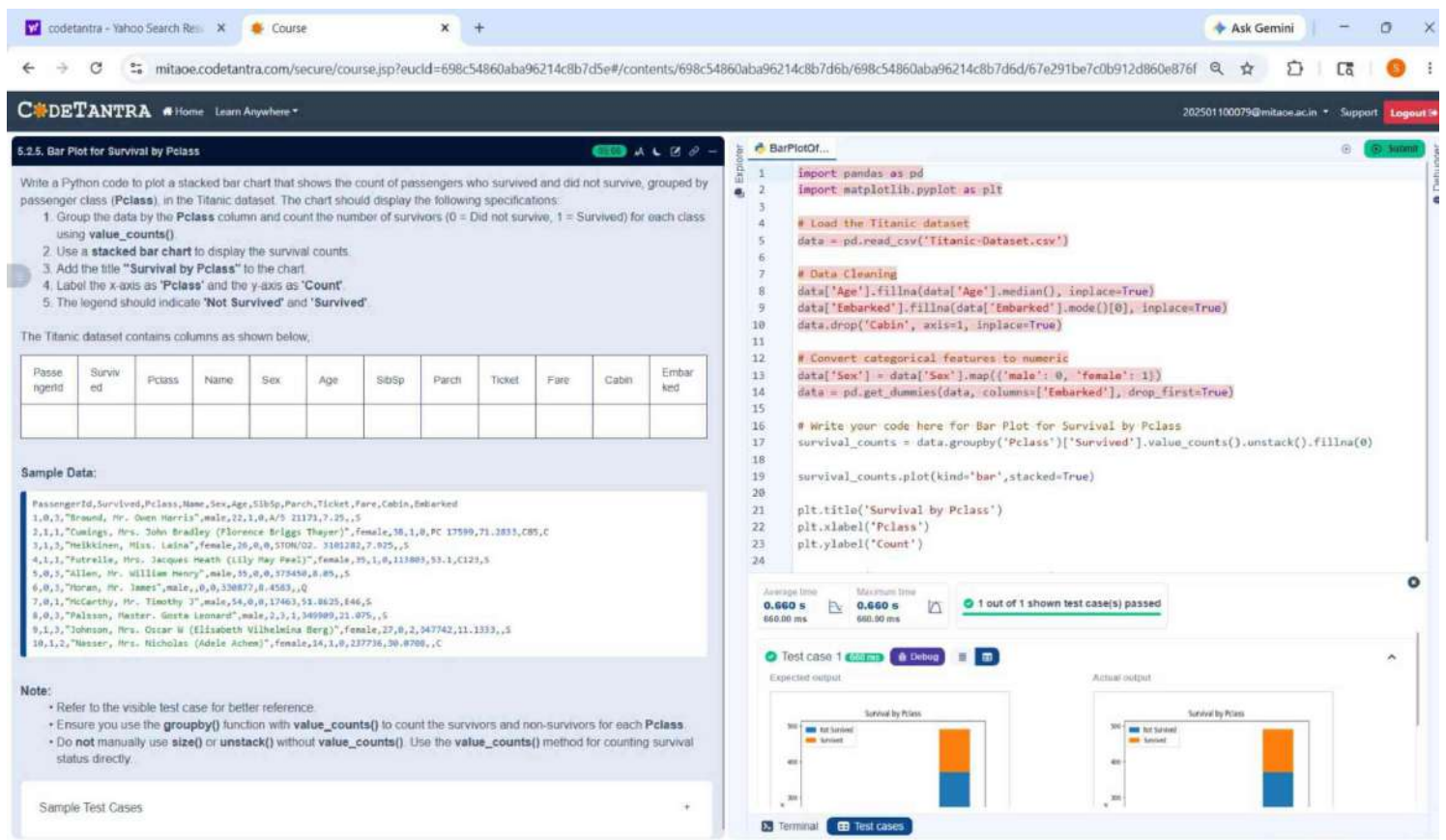
Test case 1

Expected output

Actual output

City sold the most products: Los Angeles

City sold the most products: Los Angeles



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5.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use 30 bins for the histogram
2. Set the edge color of the bars to black (k).
3. Label the x-axis as 'Age' and the y-axis as 'Frequency'.
4. Add the title "Age Distribution" to the histogram.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 11171	7.25	S	
2	1	1	Cutings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	FC 17599	71.2833	C85	C
3	1	3	Helonen, Miss. Laina	female	16	0	0	STON/O 3101327	5.12		S
4	1	1	Kutvikk, Mrs. Jacobus Veer (Ellen Bay West)	female	36	1	0	113583	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	37460	8.66		S
6	0	3	Moran, Mr. James	male	0	0	0	104377	0.4683		Q
7	0	1	McCarthy, Mr. Timothy J	male	34	0	0	17403	51.8625	E46	S
8	0	3	Malison, Master. Gusta Leonard	male	2	1	1	24099	11.875		S
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	17	0	2	347942	31.1333		S
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	34	1	0	277736	36.8708		C

Sample Data:

```

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,Braund, Mr. Owen Harris,male,22,1,0,A/5 11171,7.25,S
2,1,1,Cutings, Mrs. John Bradley (Florence Briggs Thayer),female,38,1,0,FC 17599,71.2833,C85,C
3,1,3,Helonen, Miss. Laina,female,16,0,0,STON/O 3101327,5.12,S
4,1,1,Kutvikk, Mrs. Jacobus Veer (Ellen Bay West),female,36,1,0,113583,53.1,C123,S
5,0,3,Allen, Mr. William Henry,male,35,0,0,37460,8.66,S
6,0,3,Moran, Mr. James,male,0,0,0,104377,0.4683,Q
7,0,1,McCarthy, Mr. Timothy J,male,34,0,0,17403,51.8625,E46,S
8,0,3,Malison, Master. Gusta Leonard,male,2,1,1,24099,11.875,S
9,1,3,Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg),female,17,0,2,347942,31.1333,S
10,1,2,Nasser, Mrs. Nicholas (Adele Achem),female,34,1,0,277736,36.8708,C
  
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

+

Histogram

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male':0, 'female':1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Histogram
17 plt.hist(data['Age'], bins=30, edgecolor='k')
18 plt.title('Age Distribution')
19 plt.xlabel('Age')
20 plt.ylabel('Frequency')
21 plt.show()
  
```

Download Time

0.330 s

560.00 ms

Upload Time

0.350 s

560.00 ms

1 out of 1 shown test case(s) passed

Test case 1

Debug

Test

Expected output

Actual output

Terminal

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3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using `numpy` based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases

customs...

```

1 import numpy as np
2
3 # Take user input for the start, stop, and step of the sequence
4 start = int(input())
5 stop = int(input())
6 step = int(input())
7
8 # Generate the sequence using np.arange()
9 sequence = np.arange(start, stop, step)
10 # Print the generated sequence
11 print(sequence)

```

Average time

0.012 s

11.75 ms

Maximum time

0.016 s

16.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

Actual output

[1 3 7 9]

[1 3 7 9]

Terminal

Test cases

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1.2.6. Numpy: Searching, Setting, Counting, Broadcasting

The given code in the editor takes a single array, `array1`, as space-separated integers as input from the user. Additionally, it takes the following inputs:

- `search_value`: The value to search for in the array.
- `count_value`: The value to count its occurrences in the array.
- `broadcast_value`: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- Searching**: Find the indices where `search_value` appears in `array1` and print these indices.
- Counting**: Count how many times `count_value` appears in `array1` and print the count.
- Broadcasting**: Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
- Sorting**: Sort `array1` in ascending order and print the sorted array.

Input Format:

- A single line containing space-separated integers representing `array1`.
- An integer `search_value` represents the value to search for in the array.
- An integer `count_value` represents the value to count in the array.
- An integer `broadcast_value` represents the value to add to each element of the array.

Output Format:

- The indices where `search_value` occurs in `array1`.
- The count of occurrences of `count_value` in `array1`.
- The array after adding the `broadcast_value` to each element.
- The sorted array.

Sample Test Cases

arrayOps.py

```

1 import numpy as np
2
3 # Input array from the user
4 array1 = np.array(list(map(int, input().split())))
5
6 # Searching
7 search_value = int(input("Value to search: "))
8 count_value = int(input("Value to count: "))
9 broadcast_value = int(input("Value to add: "))
10
11 # Find indices where value matches in array1
12 a = np.where(array1==search_value)[0]
13 print(a)
14
15 # Count occurrences in array1
16 b = np.count_nonzero(array1==count_value)
17 print(b)
18
19 # Broadcasting addition
20 c = array1 + broadcast_value
21 print(c)
22
23 # Sort the first array
24 d = np.sort(array1)
25 print(d)

```

Average time: 0.016 s

Maximum time: 0.019 s

16.00 ms

16.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

1 1 1 2 2

Value to search: 1

Value to count: 1

Value to add: 1

[0 1 2]

5

Actual output

1 1 1 2 2

Value to search: 1

Value to count: 1

Value to add: 1

[0 1 2]

5

Terminate

Test cases

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3.2.1. Numpy: Matrix Operations

The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. Addition (`matrix_a + matrix_b`)
2. Subtraction (`matrix_a - matrix_b`)
3. Element-wise Multiplication (`matrix_a * matrix_b`)
4. Matrix Multiplication (`matrix_a * matrix_b`)
5. Transpose of Matrix A

Input Format:

- The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

Sample Test Cases

```

1 import numpy as np
2
3 # Input matrices
4 print("Enter Matrix A:")
5 matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
6
7 print("Enter Matrix B:")
8 matrix_b = np.array([list(map(int, input().split())) for i in range(3)])
9
10
11 # Addition
12 s_result = matrix_a + matrix_b
13 print("Addition (A+B):")
14 print(s_result)
15
16 # Subtraction
17 s_result = matrix_a - matrix_b
18 print("Subtraction (A-B):")
19 print(s_result)
20
21 # Multiplication (element-wise)
22 m_result = matrix_a * matrix_b
23 print("Element-wise Multiplication (A * B):")
24 print(m_result)
25
26 # Matrix multiplication (dot product)
27 matrix_multiplication = np.dot(matrix_a, matrix_b)
28 print("A dot B:")
29 print(matrix_multiplication)
30
31 # Transpose
32 transpose_a = matrix_a.T
33 print("Transpose of A:")
34 print(transpose_a)

```

Average time

0.021 s

25.08 ms

Maximum time

0.029 s

29.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

Actual output

Enter Matrix A:

Enter Matrix A:

Enter Matrix B:

Enter Matrix B:

Terminal

Test cases

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4.1.2. Dictionary to dataframe

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column Gender with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the Age column.
- Display the DataFrame after deleting the column.

Sample Test Cases

```

1 import pandas as pd
2
3 # Provided dictionary of lists
4 data = {
5     'Name': ['Alice', 'Bob', 'Charlie'],
6     'Age': [25, 30, 35]}
7
8
9 # Convert the dictionary to a DataFrame
10 df = pd.DataFrame(data)
11
12 # Display the original DataFrame
13 print("Original DataFrame:")
14 print(df)
15
16 # Adding a new row
17 new_name = input("New name: ")
18 new_age = int(input("New age: "))
19 df.loc[len(df)] = [new_name, new_age]
20
21 # Display the DataFrame after adding a new row
22 print("After adding a row:\n", df)
23
24 # Modifying a row
25 row_to_modify = int(input("Index of row to modify: "))
26 new_age_value = int(input("New age: "))
27 df.at[row_to_modify, 'Age'] = new_age_value
28 # Display the DataFrame after modifying a row
29 print("After modifying a row:")
30 print(df)
31
32 # Deleting a row
33 row_to_delete = int(input("Index of row to delete: "))
34 df = df.drop(index=row_to_delete, axis=0)

```

Average time: 0.096 s

Maximum time: 0.113 s

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1 433 ms Debug Test

Expected output

Original DataFrame:

	Name	Age
0	Alice	25
1	Bob	30
2	Charlie	35

New name:

Actual output

Original DataFrame:

	Name	Age
0	Alice	25
1	Bob	30
2	Charlie	35

New name:

Terminal

Test Cases

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1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 n=int(input())
2 for i in range(1,n+1):
3     print("*" * i)
```

Average time0.004 s4.50 msMaximum time0.008 s8.00 ms

2 out of 2 shown test case(s) passed4 out of 4 hidden test case(s) passed

Test case 1

Expected output

Actual output

Terminat

Test cases

< Prev

Reset

Solve

Next >

Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

1. Insertion – Insert a new key-value pair into the dictionary using user input.
2. Update – Update the value of an existing key using user input.
3. Deletion – Delete a specified key from the dictionary using user input.
4. Traversal – Traverse the final dictionary and display all key-value pairs.

1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.
2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.
3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.
4. Finally, traverse the dictionary and display all key-value pairs.

- All operations must be performed using dictionary methods.
- Perform deletion only if possible, leaves the dictionary unchanged.
- Refer to the visible test cases for better understanding and strictly match with the input/outputs

```

1 # Initial Dictionary with 10 predefined records
2 student = {
3     1: "Amit",
4     2: "Nitya",
5     3: "Kiran",
6     4: "Neha",
7     5: "Arjun",
8     6: "Pooja",
9     7: "Rahul",
10    8: "Sneha",
11    9: "Vikram",
12    10: "Anjali"
13 }
14 print("Original Dictionary:", student)
15 key1 = int(input())
16 value1 = input()
17 student.update({key1: value1})
18 print("After Insertion:", student)
19
20 key2 = int(input())
21 value2 = input()
22 if (key2 in student.keys()):
23     student.update({key2: value2})
24 print("After Update:", student)
25
26 key3 = int(input())
27 if (key3 in student.keys()):
28     student.pop(key3)
29 print("After Deletion:", student)
30
31 print("Traversing Dictionary:")
32 for key, value in student.items():
33     print("{} [key] : (value)".format(key))

```

Expected output	Actual output
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14
15	15
16	16
17	17
18	18
19	19
20	20
21	21
22	22
23	23
24	24
25	25
26	26
27	27
28	28
29	29
30	30
31	31
32	32
33	33
34	34
35	35
36	36
37	37
38	38
39	39
40	40
41	41
42	42
43	43
44	44
45	45
46	46
47	47
48	48
49	49
50	50
51	51
52	52
53	53
54	54
55	55
56	56
57	57
58	58
59	59
60	60
61	61
62	62
63	63
64	64
65	65
66	66
67	67
68	68
69	69
70	70
71	71
72	72
73	73
74	74
75	75
76	76
77	77
78	78
79	79
80	80
81	81
82	82
83	83
84	84
85	85
86	86
87	87
88	88
89	89
90	90
91	91
92	92
93	93
94	94
95	95
96	96
97	97
98	98
99	99
100	100

```

Pawla , 7;  Nana , 8;  Soma , 1;  Einar , 10;  Asjall , 11;  Sarnak }

```

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4.2.4. Most Frequently Sold Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pairs that were sold most frequently.

Sample Data:

```

Date,Product,Quantity,Price,City
2025-01-01,Product A,5,30,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,6,20,Los Angeles
2025-01-04,Product C,5,30,New York
2025-01-05,Product B,5,15,Los Angeles
2025-01-05,Product A,2,20,Chicago
2025-01-05,Product C,3,30,Los Angeles

```

Explanation:

Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C
- 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product B
- 2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

- Product A and Product B: Appear in transactions on 2025-01-01 and 2025-01-03.
- Product A and Product C: Appear in transactions on 2025-01-02 and 2025-01-05.
- Product B and Product C: Appear in transactions on 2025-01-04.

Most Frequent Product Combinations:

- Product A and Product B (2 times)
- Product A and Product C (2 times)

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

4.2.4. Most Frequently Sold Product Pairs

```

1 # Import pandas as pd
2 # from itertools import combinations
3 # from collections import Counter
4
5 # Prompt user to input the file name
6 file_name = input()
7
8 # Read data from the specified CSV file
9 df = pd.read_csv(file_name)
10
11
12 # Output the most frequent product pairs
13
14 import pandas as pd
15 from itertools import combinations
16 from collections import Counter
17
18 # Prompt user to input the file name
19 file_name = input()
20
21 # Read data from the specified CSV file
22 df = pd.read_csv(file_name)
23
24 # write the code
25 grouped = df.groupby("Date")["Product"].apply(list)
26
27 pairs_counter = Counter()
28 for products in grouped:
29     pairs = combinations(sorted(products), 2)
30     pairs_counter.update(pairs)
31 max_frequency = max(pairs_counter.values())
32 most_common_pairs = [(pair, freq) for pair, freq in pairs_counter.items() if freq == max_frequency]
33 # Output the most frequent product pairs
34 for (product1, product2), frequency in most_common_pairs:

```

Average time: 0.018 s
Maximum time: 0.037 s

1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1

Expected output	Actual output
Product A and Product B: 2 times	Product A and Product B: 2 times
Product A and Product C: 2 times	Product A and Product C: 2 times

Terminal Test cases

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

[illegible]

PassengerID,Survived,Class,Sex,Age,SibSp,ParCh,Ticket,Fare,Cabin,Embarked
1,0,1,"Brandt, Mr. Owen Harris",male,22,1,0,A/5 21171-7,25.,5
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 35969,71.2835,C85,C
3,1,0,"Heikkinen, Miss. Laina",female,26,0,STON/O2 3101282,7.925,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,5
5,0,2,"Allen, Mr. William Henry",male,35,0,37450,0.85,5
6,0,2,"Hansen, Mr. Joes",male,0,0,10067,0.4385,9
7,0,2,"McCarthy, Mrs. Timothy J.",male,55,0,17463,51.86,5
8,0,2,"Palsson, Master. Gosta Leonard",male,2,1,1,349929,21.66,5
9,1,2,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,547742,51.1333,5
10,1,2,"Nasser, Mr. Nicholas (Adèle Achmet)",female,14,1,0,23776,30.0706,C

Sample Test Cases

© SAGE Publications

Average time: 0.310 s 310.00 ms Maximum time: 0.310 s 310.00 ms 1 out of 1 shown test case(s) passed

Test case 1 3/15/2020 Debug ⌵ ⌵

Expected output



Actual output



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4.1.1. Pandas - series creation and manipulation

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

```

1 import pandas as pd
2
3 # Take inputs from the user to create a list of numbers
4 numbers = list(map(int, input().split()))
5
6 # Create a Pandas series from the list of numbers
7 series = pd.Series(numbers)
8 # Grouping by even and odd numbers and calculating the mean
9 grouped = series.groupby(series % 2 == 0).mean()
10
11 # Display the mean of even and odd numbers with labels
12 grouped.index = ['even' if i % 2 == 0 else 'odd' for i, even in grouped.index]
13 print("Mean of even and odd numbers:")
14 print(grouped)
15

```

Average time: 0.012 s

Maximum time: 0.022 s

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1

Expected output:

```

1 2 3 4 5 6 7 8
Mean of even and odd numbers:
odd    5.0
even    6.0
dtype: float64

```

Actual output:

```

1 2 3 4 5 6 7 8
Mean of even and odd numbers:
odd    5.0
even    6.0
dtype: float64

```

Terminar

Test cases

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2.1.1. List operations

Write a Python program that implements a menu-driven interface for managing a list of integers. The program should have the following menu options:

1. Add

2. Remove

3. Display

4. Quit

The program should repeatedly prompt the user to enter a choice from the menu. Depending on the choice selected, the program should perform the following actions:

- **Add:** Prompts the user to enter an integer and add it to the integer list. If the input is not a valid integer, display "Invalid input".
- **Remove:** Prompts the user to enter an integer to remove from the list. If the integer is found in the list, remove it, otherwise, display "Element not found". If the list is empty, display "List is empty".
- **Display:** Displays the current list of integers. If the list is empty, display "List is empty".
- **Quit:** Exits the program.
- The program should handle invalid menu choices by displaying "Invalid choice". Ensure that the program continues to prompt the user until they choose to quit (option 4).

Sample Test Cases

listOps.py

```

1 list=[]
2 while True:
3     print("1. Add")
4     print("2. Remove")
5     print("3. Display")
6     print("4. Quit")
7     a=int(input("Enter choice: "))
8     if a==1:
9         v=input("Integer: ")
10        if v.isdigit()==True:
11            list.append(int(v))
12            print("List after adding:",list)
13        else:
14            print("Invalid input")
15    elif a==2:
16        if len(list)==0:
17            print("List is empty")
18        else:
19            f=int(input("Integer: "))
20            if f in list:
21                list.remove(int(f))
22                print("List after removing:",list)
23            else:
24                print("Element not found")
25    elif a==3:
26        if len(list)==0:
27            print("List is empty")
28        else:
29            print(list)
30    elif a==4:
31        break
32    else:
33        print("Invalid choice")
34

```

Average time

0.077 s

16.47 ms

Maximum time

0.103 s

103.04 ms

2 out of 2 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1

Expected output

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 1

Integer: 12

Actual output

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 1

Integer: 12

Terminal

Test cases

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1.1.2. Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:
 $1 \leq n \leq 999$

Input Format:
 The input consists of a single integer n .

Output Format:
 If n is a single-digit number, print its square.
 If n is a two-digit number, print its square root (rounded to two decimal places).
 If n is a three-digit number, print its cube root (rounded to two decimal places).
 Else print "Invalid"

Sample Test Cases

condition...

```

1 import math
2 n=int(input(""))
3 if(n>0 and n<=9):
4     print(n*n)
5 elif(n>=10 and n<=99):
6     print(f"{math.sqrt(n):.2f}")
7 elif(n>=100 and n<=999):
8     print(f"{n**(1/3):.2f}")
9 else:
10    print("Invalid")

```

Average time
0.006 s
5.88 ms

Maximum time
0.007 s
7.00 ms

4 out of 4 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1
Debug

Expected output	Actual output
81	81

Test case 2
Test case 3

Terminal
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5.2.1. Titanic Dataset

Dataset Information:

- **Polars:** Passenger class (1 = First, 2 = Second, 3 = Third)
- **Gender:** Gender of the passenger (male/female).

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns)

Bar Plot (Pclass Distribution)

- Create a bar plot to show the distribution of passengers across the different passenger classes (Pclass).

- Use the color skyblue for the bars.
- Title the plot as "Passenger Class Distribution".

- Create a pie chart to display the distribution of male and female passengers.

- Use lightblue for males and lightcoral for females.

- Histogram (Age Distribution):

- Create a histogram to visualize the distribution of passengers' ages.
- Use lightgreen for the bars with black edges (edgecolor = 'black').

- Bar Plot (Survival Count):

- Create a bar plot to show the count of passengers who survived and those who did not, based on the Survived column
- Use the colors lightblue for survivors (1) and lightcoral for non-survivors (0)

- Scatter Plot (Fare vs Age)

- Create a scatter plot to visualize the relationship between the Fare and Age of passengers.

- Label the x-axis as "Age" and the y-axis as "Fare".

Export results as .id

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset from the CSV file
5 df = pd.read_csv('titanic.csv')
6
7 # Set up the figure for 3 subplots
8 fig, axes = plt.subplots(3, 2, figsize=(12, 12))
9
10 # write the code..
11
12 #plot 1: Count of passenger by class
13 axes[0,0].bar(df['Pclass'].value_counts().index,
14              df['Pclass'].value_counts(), color='skyblue')
15 axes[0,0].set_title("Passenger Class Distribution")
16 axes[0,0].set_xlabel("Pclass")
17 axes[0,0].set_ylabel("Count")
18
19 #plot 2: Gender Distribution
20 axes[0,1].pie(df['gender'].value_counts(), labels=df['gender'].value_counts().index, autopct='%1.1f%%', color=
21              ['lightblue', 'lightcoral'])
22 axes[0,1].set_title("Gender Distribution")
23
24 #plot 3: Age Distribution
25 axes[1,0].hist(df['Age'].dropna(), bins=8, color='lightgreen', edgecolor='black')
26 axes[1,0].set_title("Age Distribution")
27 axes[1,0].set_xlabel("Age")
28 axes[1,0].set_ylabel("frequency")
29
30 #plot 4: Survival Count
31 axes[1,1].bar(df['survived'].value_counts().index, df['survived'].value_counts(), color=['lightblue', 'lightcoral'])
32 axes[1,1].set_title("Survival Count")
33 axes[1,1].set_xlabel("Survived (0 = No, 1 = Yes)")
34
35 axes[2,0].set_xlabel("Count")

```

1250.00 mva	1250.00 mva	
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Expected output	Actual output
-----------------	---------------

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer.

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 n=int(input(""))
2 sum=0
3 while(n>0);
4     rem=n%10
5     sum=sum*10+rem
6     n=n//10
7 print(sum)
```

Average time

0.004 s

4.40 ms

Maximum time

0.007 s

7.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1

Expected output

5367

7635

Actual output

5367

7635

Test case 2

Terminal

Test cases

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4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.

2. Convert the 'Sex' column to numeric values (male: 0, female: 1).

3. One-hot encode the 'Embarked' column, dropping the first category.

4. Get the mean age of passengers.

5. Get the median fare of passengers.

6. Get the number of passengers by class.

7. Get the number of passengers by gender.

8. Get the number of passengers by survival status.

9. Calculate the survival rate of passengers.

10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Owen Morris	male	22	1	0	A/5 21171	53.1		S
2	1	1	Mr. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Mr. James Watson	male	32	0	0	17423	51.0		S
4	1	1	Mr. William Henry	male	30	0	0	17554	51.0		S
5	0	3	Mr. James Watson	male	32	0	0	17423	51.0		S
6	1	1	Mr. William Henry	male	30	0	0	17554	51.0		S
7	0	3	Mr. James Watson	male	32	0	0	17423	51.0		S
8	1	1	Mr. William Henry	male	30	0	0	17554	51.0		S
9	0	3	Mr. James Watson	male	32	0	0	17423	51.0		S
10	1	1	Mr. William Henry	male	30	0	0	17554	51.0		S

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1, 0, 3, Mr. Owen Morris, male, 22, 1, 0, A/5 21171, 53.1, , S

2, 1, 1, Mrs. John Bradley (Florence Briggs Thayer), female, 38, 1, 0, PC 17599, 71.2833, C85, C

3, 1, 3, Mr. James Watson, male, 32, 0, 0, 17423, 51.0, , S

4, 1, 1, Mr. William Henry, male, 30, 0, 0, 17554, 51.0, , S

5, 0, 3, Mr. James Watson, male, 32, 0, 0, 17423, 51.0, , S

6, 1, 1, Mr. William Henry, male, 30, 0, 0, 17554, 51.0, , S

7, 0, 3, Mr. James Watson, male, 32, 0, 0, 17423, 51.0, , S

8, 1, 1, Mr. William Henry, male, 30, 0, 0, 17554, 51.0, , S

9, 0, 3, Mr. James Watson, male, 32, 0, 0, 17423, 51.0, , S

10, 1, 1, Mr. William Henry, male, 30, 0, 0, 17554, 51.0, , S

Note: Refer to the visible test case for better reference.

Sample Test Cases

TitanicDataset

1

import pandas as pd

2

import numpy as np

3

4

Load the Titanic dataset

5

data = pd.read_csv('Titanic-Dataset.csv')

6

data['FamilySize'] = data['SibSp'] + data['Parch']

7

8

1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)

9

data['IsAlone'] = (data['FamilySize'] == 0).astype(int)

10

11

2. Convert 'Sex' to numeric (male: 0, female: 1)

12

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

13

14

3. One-hot encode the 'Embarked' column

15

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

16

17

4. Get the mean age of passengers

18

mean_age = data['Age'].mean()

19

20

21

5. Get the median fare of passengers

22

median_fare = data['Fare'].median()

23

24

25

6. Get the number of passengers by class

26

passenger_by_class = data['Pclass'].value_counts()

27

28

29

7. Get the number of passengers by gender

30

passenger_by_gender = data['Sex'].value_counts()

31

32

33

8. Get the number of passengers by survival status

34

passenger_by_survival = data['Survived'].value_counts()

Run time: 0.007 s

Maximum time: 0.007 s

1 out of 1 shown test case(s) passed

Test case 1

Expected output

29.40111747019402

14.4542

0 - 401

1 - 214

2 - 284

Name: Pclass, dtype: int64

Actual output

29.40111747019402

14.4542

0 - 401

1 - 214

2 - 284

Name: Pclass, dtype: int64

Terminated

Test cases

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4.2.5. Titanic Dataset Analysis and Data Cleaning

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.

2. Display the last 5 rows of the dataset.

3. Get the shape of the dataset (number of rows and columns).

4. Get a summary of the dataset (using .info()).

5. Get basic statistics (mean, standard deviation, etc.) of the dataset using .describe().

6. Check for missing values and display the count of missing values for each column.

7. Fill missing values in the 'Age' column with the median age.

8. Fill missing values in the 'Embarked' column with the most frequent value (mode()).

9. Drop the 'Cabin' column due to many missing values.

10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below.

PassengerID	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerID, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,1,"Braund, Mr. Owen Harris",male,22,1,0,0,1,53.1,1,5
2,1,1,"Cummings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17596,71.283,C85,C
3,1,1,"Heikkinen, Miss. Laina",female,26,0,0,170001,7.25,0
4,1,1,"Hutera, Mrs. Jacques Heath (Lily May Peel)",female,29,1,0,12109,53.1,C123,S
5,0,1,"Allen, Mr. William Henry",male,35,0,0,17500,8.45,0
6,0,1,"Moran, Mr. James",male,0,0,108077,0.4882,0
7,0,1,"McCarthy, Mr. Timothy J",male,34,0,0,17422,51.002,146,S
8,0,1,"Palsson, Master. Gosta Leonard",male,2,3,1,40980,21.015,0
9,1,1,"Johnson, Mrs. Oscar W (Ellenath Vilhelmina Berg)",female,27,0,1,247702,51.1232,5
10,1,1,"Nasser, Mrs. Ninales (Abdie Schen)",female,14,1,0,137736,30.0760,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

TitanicData.py

1

2

3

4

5

6

7

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16

17

18

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20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

Import pandas as pd

Import numpy as np

Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

import pandas as pd

import numpy as np

Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

1. Display the first 5 rows of the dataset

print(data.head())

2. Display the last 5 rows of the dataset

print(data.tail())

3. Get the shape of the dataset

print(data.shape)

4. Get a summary of the dataset (info)

print(data.info())

5. Get basic statistics of the dataset

print(data.describe())

6. Check for missing values

print(data.isnull().sum())

7. Fill missing values in the 'Age' column with the median age

data['Age'].fillna(data['Age'].median(), inplace=True)

Execution time: 0.318 s

Maximum time: 0.318 s

1 out of 1 shown test case(s) passed

Test Case 1

Expected output

Actual output

PassengerID Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked

0 1 0 1 Braund, Mr. Owen Harris male 22 1 0 0 1 53.1 1 5

1 1 1 1 Cummings, Mrs. John Bradley (Florence Briggs Thayer) female 38 1 0 PC 17596 71.283 C85 C

2 1 1 1 Heikkinen, Miss. Laina female 26 0 0 170001 7.25 0

3 1 1 1 Hutera, Mrs. Jacques Heath (Lily May Peel) female 29 1 0 12109 53.1 C123 S

4 0 1 1 Allen, Mr. William Henry male 35 0 0 17500 8.45 0

5 0 1 1 Moran, Mr. James male 0 0 108077 0.4882 0

6 0 1 1 McCarthy, Mr. Timothy J male 34 0 0 17422 51.002 146 S

7 0 1 1 Palsson, Master. Gosta Leonard male 2 3 1 40980 21.015 0

8 1 1 1 Johnson, Mrs. Oscar W (Ellenath Vilhelmina Berg) female 27 0 1 247702 51.1232 5

9 1 1 1 Nasser, Mrs. Ninales (Abdie Schen) female 14 1 0 137736 30.0760 C

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Test Cases

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5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the 'Sex' column, then use the value_counts() function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.

2. Use a stacked bar chart to display the survival counts.

3. Add the title "Survival by Gender" to the chart.

4. Label the x-axis as 'Gender' and the y-axis as 'Count'.

5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	
2	1	1	Cooking, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101283	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	97450	8.05	S	
6	0	3	Nolan, Mr. James	male	0	0	0	520877	8.4583	Q	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palston, Master. Gosta Leonard	male	2	1	1	340909	21.075	S	
9	1	3	Johnson, Mrs. Oscar W (Elizabeth Vilhelmina Berg)	female	27	0	2	347742	11.1333	S	
10	1	2	Bassett, Mrs. Nicholas (Adele Achen)	female	14	1	0	237736	30.0708	C	

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cooking, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101283,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,97450,8.05,S
6,0,3,"Nolan, Mr. James",male,,0,0,520877,8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palston, Master. Gosta Leonard",male,2,1,1,340909,21.075,S
9,1,3,"Johnson, Mrs. Oscar W (Elizabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S
10,1,2,"Bassett, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Gender
17 survival_counts = data.groupby('Sex')['Survived'].value_counts().unstack()
18 survival_counts.plot(kind='bar', stacked=True)
19
20 plt.title('Survival by Gender')
21 plt.xlabel('Gender')
22 plt.ylabel('Count')
23 plt.legend(['Not Survived', 'Survived'])
24 plt.show()
```

Average time: 0.521 s

Maximum time: 0.521 s

1 out of 1 shown test case(s) passed

Test case 1 451 ms

Debug

Expected output

Actual output

Survival by Gender

Survival by Gender

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1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format
`<number> x <multiplier> = <result>`

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

1n= int(input(""))2for i in range(1,11):3 print(n, 'x', i, "=", n*i)45

Average time0.007 s6.79 msMaximum time0.012 s12.99 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1Expected output8 x 1 = 88 x 2 = 168 x 3 = 248 x 4 = 328 x 5 = 40Actual output8 x 1 = 88 x 2 = 168 x 3 = 248 x 4 = 328 x 5 = 40

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Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

The Titanic dataset contains columns as shown below.

Sample Data:

Note: Refer to the visible test case for better reference

Sample Test Cases

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival Rate
17 survival_counts = data['Survived'].value_counts()
18 survival_counts.plot(kind='bar')
19 plt.title('Survival Count')
20 plt.xlabel('Survived')
21 plt.ylabel('Count')
22 plt.show()
23

```

Average time: 0.367 s. Maximum time: 0.367 s. for 1 out of 1 shown test case(s) passed

Test case 1



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3.2.2. NumPy: Horizontal and Vertical Stacking of Arrays

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking: Stack the two matrices horizontally (side by side).
- Vertical Stacking: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

```
1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])
6
7 print("Enter Array2:")
8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])
9
10 # Perform horizontal stacking (hstack)
11 hstack_result = np.hstack((arr1, arr2))
12
13
14 # Perform vertical stacking (vstack)
15 vstack_result = np.vstack((arr1, arr2))
16 print("Horizontal Stack:")
17 print(hstack_result)
18 print("Vertical Stack:")
19 print(vstack_result)
```

Average time: 0.021 s
Maximum time: 0.025 s
21.00 ms 20.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1

Expected output

Enter Array1:

1 2 3
4 5 6
7 8 9

Enter Array2:

1 2 3
4 5 6
7 8 9

Actual output

Enter Array1:

1 2 3
4 5 6
7 8 9

Enter Array2:

1 2 3
4 5 6
7 8 9

Test cases

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3.2.5. NumPy: Copying and Viewing Arrays

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

```
Original array after modifying view: original_array
View array: view_array
```

- After modifying the copy:

```
Original array after modifying copy: original_array
Copy array: copy_array
```

Sample Test Cases

copyAnd...

```

1 import numpy as np
2
3 inputlist = list(map(int, input().split(" ")))
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array = original_array.view()
10
11 # Create a copy
12 copy_array = original_array.copy()
13
14 # Modify the view
15 view_array[0] = 99
16 print("Original array after modifying view:", original_array)
17 print("View array:", view_array)
18
19 # Modify the copy
20 copy_array[1] = 88
21 print("Original array after modifying copy:", original_array)
22 print("Copy array:", copy_array)
23

```

0.005 s

5.33 ms

0.008 s

8.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

```

99 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [20 30 40 50 60 70 80]

```

Actual output

```

99 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [20 30 40 50 60 70 80]

```